

UPLB IdSC FaNS Knowledge Hub: A Knowledge Management System for Food and Nutrition Security

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Abstract—Food and nutrition security research in the Philippines is scattered across different institutions, databases, and studies, with no one unifying system to bring it all together. This makes it harder for researchers, policymakers, and even the general public to find and use the information that is already there. This study presents the UPLB IdSC-FaNS Knowledge Hub, a web-based Knowledge Management System (KMS) built specifically for the UPLB Interdisciplinary Studies Center on Food and Nutrition Security. The system was developed using a hybrid MongoDB and Elasticsearch architecture for fast full-text search and aggregations, a community layer with organizations and a moderated approval pipeline, and AI-assisted curation through Gemini-based metadata extraction and a related-insights panel that synthesizes themes and research gaps from each post. The system was evaluated through a fully remote, unmoderated study on $N = 26$ participants from the UPLB community, split across three tracks: 20 standard users, 5 organization administrators (one of whom was excluded from admin-gated analyses due to a post-hoc-identified provisioning gap, leaving $n = 4$), and 1 website administrator reported as a single-case feasibility observation. The pooled SUS came out to 83.17 (95% CI [76.35, 89.52]), placing the system in the Good–Excellent band. Twenty-two of twenty-three tasks scored $\geq 95\%$ success at $SEQ \leq 1.5$ on a 1–5 scale where 1 is easiest, and per-field AI extraction accuracy ranged from 86.4% on Year/Journal to 100% on Title and Abstract. AI metadata extraction was also the most-named useful feature in the open-ended responses. The five research objectives were all met at both the satisfaction and task-success layers, and what is left is mostly a focused bug-fix list (cite-button discoverability, post-submit server validation, sort and filter regressions, and an AI-trust dip addressable through provenance cues) rather than a structural rework.

Index Terms—knowledge management system, food security, nutrition, Elasticsearch, AI-assisted curation, metadata extraction, usability evaluation, SUS

I. INTRODUCTION

A. Background of the Study

Food security and nutrition are essential topics relevant to millions of people especially in a developing country like the Philippines. The Philippine nutrition survey in 2013 shows that over half of the more than twenty thousand participants had inadequate amounts of essential nutrients, with deficiencies in key micronutrients like Iron, Calcium, and Vitamin C affecting 90% of the population. It was also noted that participants in lower socioeconomic groups were more likely to be deficient in the essential nutrients [1]. Food insecurity, on another

hand, is disproportionately low especially after the COVID-19 pandemic among small scale farmers who were most affected by the disrupted market, larger households who share limited resources with more people, and indigenous people who have limited access to economic and social service [2].

Research on nutrition is important because it provides an understanding of how nutrition influences overall well-being, aids in disease prevention and management, and addresses the disparity of health among different socioeconomic groups [3]. These are also some of the reasons why the Interdisciplinary Studies Center on Food and Nutrition Security or IdSC-FaNS was formed. It was created under the University of the Philippines Los Baños Office of the Vice Chancellor for Research and Extension for the continued stewardship of nutrition and food security research in the country [4].

B. Statement of the Problem

Data and research on food and nutrition security in the Philippines are distributed across different databases, studies, and institutions with no one unifying system. This can hamper and delay policy makers, other researchers, and the general public in accessing and interpreting information [5]. The lack of access to comprehensive data and analytics for food security and nutrition research can lead to delayed action from policy makers, inadequate public awareness, and limited research collaboration.

This study proposes the development of a Knowledge Management System (KMS) for food and nutrition security research in the Philippines. This aims to create a centralized, accessible platform for information essential to food security research in the Philippines.

C. Significance of the Study

The Knowledge Management System (KMS) provides the IdSC-FaNS a platform for sharing knowledge on food and nutrition security. Data visualization and analytics on the current trends in food security research enhance the ability of IdSC-FaNS to monitor, interpret, and address the challenges in food and nutrition security. The use of advanced data visualization and analysis allows the IdSC-FaNS and researchers to identify trends and emerging issues in the field. This supports data-driven decisions, but also encourages targeted research that addresses the current and future issues. The KMS also provides a repository where research documents, reports, reviews, and data can be found. All of the documents are

indexed for easy retrieval and access using Elasticsearch. This helps researchers, policymakers, and other stakeholders access important information without navigating multiple sources, ultimately making research on food and nutrition security more accessible and actionable. Beyond storing and searching, the KMS introduces two AI-assisted features that increase its significance. A Gemini-based metadata extractor was added to reduce the effort of researchers in inputting research and to decrease the inaccuracy of the data within the system. A Gemini-generated *related insights* panel synthesizes themes and research gaps given the current topic of the post, showing the unseen connections that keyword search alone cannot reveal. Both features support the thesis's broader contribution: a domain-focused knowledge management system that integrates full-text search, community discussion, and AI-assisted curation in a unified platform appropriate for a developing-country research context.

D. Objectives

The main objective of this study is to develop a web application that serves as a Knowledge Management System to provide easy access to food and nutrition security research data. Specifically, it aims to:

- 1) Provide a platform for research on food and nutrition security where the public can access curated and approved research.
- 2) Enable advanced search functionality for easy access to research data using Elasticsearch.
- 3) Provide data visualization and analytics tools that help users identify trends, emerging challenges, and research gaps in food security and nutrition.
- 4) Support community-driven curation through organizations and moderated discussions.
- 5) Provide AI-assisted curation through automated metadata extraction and research insight generation.

E. Scope and Limitations

The study focuses on the design and development of a web-based Knowledge Management System for food and nutrition security in the Philippines. It collects and stores research data; it enables advanced searching and filtering; and it provides advanced data and analytic visualization. The platform relies on the availability of research data from various institutions and organizations affiliated with the ICS-FaNS.

F. Definition of Terms

- **Food Security** — a state in which all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food that meets their dietary needs and food preferences for an active and healthy life.
- **Nutrition Security** — a state where all people have access to essential nutrients and adequate diets that support physical and mental well-being, reduce the risk of disease, and promote a productive life.
- **Knowledge Management System** — a structured system that collects, stores, shares, and analyzes information within an organization.

- **Elasticsearch** — open-source search and analytics engine designed to enable fast, full-text search and data retrieval across large datasets. It is also a datastore and vector database designed for near real-time searches.

II. REVIEW OF RELATED LITERATURE

Knowledge management (KM) is the process of making the right information available to the right people. It is capturing valuable information, making it accessible, and using it to improve the work of organizations [6]. IBM identifies three types of knowledge KM usually works with namely: tacit, implicit, and explicit knowledge. Tacit knowledge comes from experience and thus is difficult to articulate and communicate. Similar to tacit knowledge, implicit knowledge comes from “know-how” knowledge that has not been documented yet. On the other hand, explicit knowledge is structured information easily articulated and documented, and can easily be transferred between people [7]. Knowledge management involves translating tacit knowledge into explicit knowledge and vice versa for easier understanding and an increase in organizational efficiency. In higher education, this can be seen as the transfer of knowledge between research and research teams, ensuring that the findings are effectively disseminated and applied.

A knowledge management system (KMS) is an information system made to facilitate the management of information in an organization. It is a tool and a platform designed to create, collect, organize, store, and disseminate knowledge to the organization. The KMS is then leveraged by the organization in order to improve decision making, and efficiency [8]. A Knowledge Management System (KMS) typically consists of several key components: databases and repositories for storing knowledge like research papers, case studies, and manuals; search engines that allow users to find specific knowledge quickly; tools that facilitate communication, such as discussion forums and shared documents; data analytics and visualization tools that help users interpret knowledge by providing insights into trends, and gaps in the research field; and knowledge sharing platforms that allow the members to contribute and share their expertise and experiences. These components make sure that knowledge is efficiently organized, accessible, and shareable within the system and the organization [9].

Several successful knowledge management systems have been developed and implemented. Harvard's DASH (Digital Access to Scholarship at Harvard) repository serves as an open-access platform where researchers can upload and share their academic work. It uses Elasticsearch for quick and accurate searching across thousands of research papers in their repository [10]. The Food and Agriculture Organization (FAO) has also developed a platform where data and research on food security are kept, and its open-access nature allows governments and policymakers from third world countries to easily view and collaborate with world-class researchers promoting innovation [11].

Despite the great effort researchers and developers have made in leveraging KMS, studies and research on KMS have not been sufficiently explored, especially in third world countries. There is still a lack of understanding on how to efficiently

utilize the knowledge in a KMS to increase efficiency in an organization [8]. However, previous research highlights diversity in focus areas; the development of a particular KMS requires specific needs appropriate to the topic and target user group of the system. Maharani et al. [8] suggests the People, Process, Technology (PPT) Framework is crucial to the design and development of the said system. The PPT framework states that the people who use the system need the right skills; the process needs to be properly structured; and the technology should be able to support both. This allows the KMS to function without resistance to change of the people and the process, as well as the technological inefficiencies to significantly affect the system.

Artificial intelligence and machine learning integration can also provide KMS enhanced functionality. Natural Language Processing or NLP can be used to analyze large amounts of text data, and extract key insights or trends related to food security. Other AI and ML tools like AI-powered search engines increase the accuracy and efficiency of the engine [12]. In this paper, a variety of statistical and algorithmic AI and ML techniques were examined for their advantages and disadvantages, most of which proved advantageous to the search engine's relevance scoring and type matching among others. Utilizing the appropriate techniques is consequential to the performance of the search engine and ultimately the KMS being developed.

III. MATERIALS AND METHODS

This chapter outlines the methodology used to design and implement a food and nutrition security research knowledge management system. The primary objective of this project is to design and develop an efficient KMS - UPLB IdSC-FaNS Knowledge Hub, that makes the access and collaboration of knowledge and ideas easier especially for students, researchers, and policymakers.

A. Technology Stack

The following technologies are used to create the web-based Knowledge Management System:

- **React** — a JavaScript library used for building dynamic user interfaces.
- **Node.js** — a JavaScript runtime environment used for backend API development and other server-side processes.
- **Next.js** — a React framework that enables server-side rendering and static site generation for building web applications.
- **Express.js** — a Node.js web framework for building RESTful APIs.
- **MongoDB** — NoSQL database used for storing structured data.
- **Elasticsearch** — a full-text search engine.
- **TanStack React Query** — an asynchronous state management library used for fetching, caching, and synchronizing server state in React apps.
- **Axios** — a promise-based HTTP client used to make requests to APIs from both the browser and Node.js.

- **Socket.io** — a library that enables real-time, bi-directional, and event-based communication between web clients and servers.
- **Tailwind CSS** — a utility-first CSS framework used for rapidly building custom user interfaces without leaving your HTML.
- **shadcn/ui** — a collection of re-usable, accessible UI components built with Tailwind CSS and Radix UI that you copy and paste into your apps.
- **Recharts** — a composable charting library built on React components, used for rendering data visualizations and graphs.
- **Google Auth** — an authentication service that allows users to sign in to applications using their existing Google account credentials.
- **Gemini 2.5 Flash** — Google's high-efficiency multi-modal AI model, optimized for speed and reasoning across text, images, and code.

TABLE I
INFRASTRUCTURE

Component	Provider	Role
App Platform (sgp1)	DigitalOcean	Frontend + backend hosting
Managed MongoDB	DigitalOcean	Production database
Managed Elasticsearch	DigitalOcean	Production search
Spaces + CDN	DigitalOcean	PDF and image storage
Gemini API	Google	AI extraction and insight generation
CrossRef REST API	CrossRef	DOI-based metadata enrichment

Table I shows the infrastructure the KMS is built on. It shows the components, the provider of that component, and what its role is. Most of the components are set up on DigitalOcean. The frontend and the backend are both on separate app platforms.

B. System Architecture

A few of the main goals of the KMS are to provide easy access to research and knowledge through search and discoverability; the system should allow quick and full-text searching through data. In order to achieve this, the application employs polyglot persistence or the use of multiple databases to support full searching capabilities while keeping storage overhead low and maintaining the integrity of the database.

The KMS uses two different technologies:

- MongoDB stores structured and relational data like Users, Organizations, and others.
- Elasticsearch handles content data, i.e. research papers, posts, and comments for quick searching and retrieval.

MongoDB stores all transactional and relational data: users, organizations, posts, comments, papers, notifications, organization-creation requests, user activities, featured posts, reports, moderation logs, and AI-insight cache. Elasticsearch stores a denormalized, projection of posts (`kms.posts`) and papers (`kms.papers`), both of which are optimized

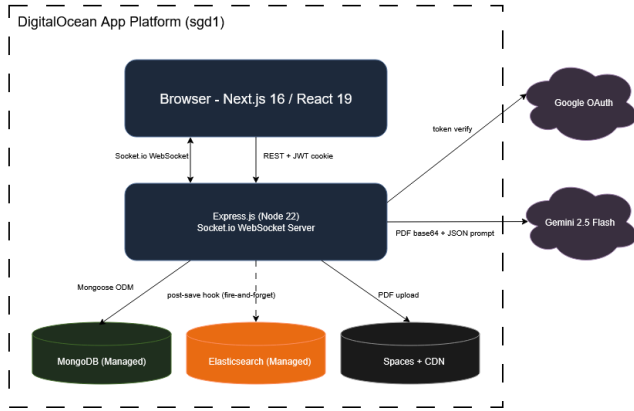


Fig. 1. System Architecture

for searching. Synchronization from MongoDB to Elasticsearch is handled by Mongoose `post('save')` and `post('findOneAndUpdate')` hooks defined directly on the Post and Paper schemas. Mongoose acts as a middleware and checks data synchronization between the two databases.

PDFs and images are stored in **DigitalOcean Spaces** (an S3-compatible object store) and delivered through the Spaces CDN. MongoDB documents persist only the CDN URL, keeping document size small.

A database schema defines how data is structured in the system. Because the KMS features a hybrid architecture with MongoDB and Elasticsearch, the database schema is labelled with two parts: one for MongoDB; one for Elasticsearch. However, both parts still function as one because they are synchronized.



Fig. 2. MongoDB-Elasticsearch Schema Connection

As shown in Figure 2, the User, and Organization, are stored and accessed in MongoDB. Posts and papers are indexed in Elasticsearch as these are the searchable data in the application.

MongoDB's text index lacks four capabilities required for advanced searching: field-level boosting (e.g., title above body), result highlighting via `<mark>`, fuzzy matching, and aggregations over full-text matches. Elasticsearch supplies all four via BM25-based relevance scoring on Apache Lucene [13], [14]. The system therefore maintains two stores and synchronizes them via Mongoose hooks.

The system is a separated frontend and backend. The Next.js frontend is compiled to a production bundle and served from DigitalOcean App Platform. It communicates with the Express backend over a REST API using Axios, with JWT session cookies set as `httpOnly`, `Secure`, and `SameSite=None` to allow cross-origin requests from the authenticated frontend. A persistent Socket.io websocket channel is open to deliver the real-time notifications.

TABLE II
ELASTICSEARCH INDICES

Index	Source	Denormalized fields	Boosted fields
kms_posts	posts	author name, organization name, tags	title ³ , body ¹ , tags ²
kms_papers	papers	first-author surname, journal	title ³ , abstract ¹ , keywords ² , authors ²

C. Key System Features

The IdSC FaNS Knowledge Hub is designed to enhance the accessibility, discovery, collaboration, and sharing of knowledge related to food and nutrition security. Below are the key features that make the app an effective platform for researchers, students, and policymakers.

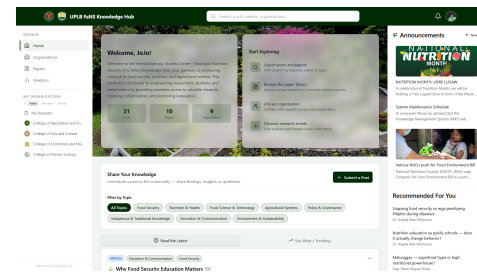


Fig. 3. Home Page

- Centralized Repository:** The system allows searching, filtering, and sorting of all published content (posts and papers). Content curation is centered around **Organizations**, which are units of moderation with a four-tier role hierarchy: owner, administrator, member, and follower. Members may submit posts to their organizations; submissions enter a pending state and become published only after an organization administrator approves them, or hidden if rejected with a reason. Once published, papers and posts are viewable to all users regardless of their role in an organization. Depending on a user's

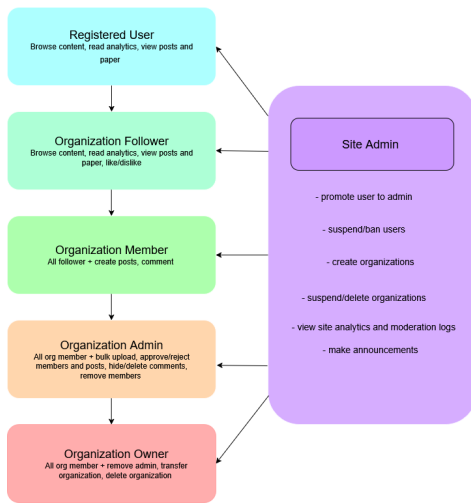


Fig. 4. Organization Hierarchy

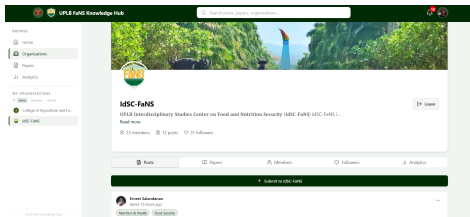


Fig. 5. Organization Page

role in the organization the content belongs to, it is also interactable (like, dislike, comment).

- 2) **Advanced Search:** The global `/search` page presents three tabs (All, Posts, Papers), a left filter panel, paginated results, and marks highlighting matching terms in the results. All searches happen through Elasticsearch with field level per Table II. The result relevance is BM25 weighted. The home page can also be filtered through eight pre-made topics and categories discussed in **Other AI Enhancements** key feature.

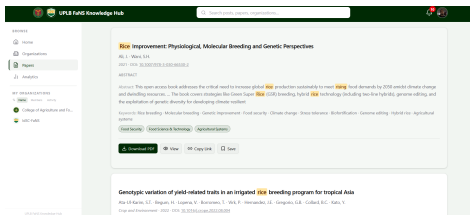


Fig. 6. Papers Page with mark

- 3) **Analytics and Data Visualization:** The admin dashboard keeps track of system-wide information and statistics on users, organizations, papers, and posts. The organization analytics tab, present in every organization page, shows posts over time, member growth, post-type distribution, and top tags of the organization. The site-wide research trends page is open to all registered users and reports research output, research areas, and organization comparisons. All charts are rendered with Recharts from data fetched from Elasticsearch aggrega-

tions on the backend.

TABLE III
ANALYTICS CHARTS (14 TOTAL)

Scope	Chart	Data source
Site	Posts over time	kms_posts aggregation
Site	Papers over time	kms_papers aggregation
Site	New users over time	users.createdAt aggregation
Site	Post-type distribution	kms_posts post_type terms agg
Site	Top tags	kms_posts tags terms agg
Site	Top organizations by post count	kms_posts organization terms agg
Org	Posts over time in the org	kms_posts filter + date histogram
Org	Member growth	organizations.members over time
Org	Post-type distribution within org	kms_posts filter + terms agg
Org	Top tags within org	kms_posts filter + terms agg
Public	Research output	kms_papers aggregation
Public	Most active research areas	kms_posts.tags terms agg
Public	Organization comparisons	kms_posts org × type matrix agg
Per-post	AI-summarized related insights	Gemini + cache

- 4) **AI Analytical Tools:** AI related insights are rendered on each post page, providing a qualitative analysis of the posts and the research surrounding the post. It provides a Gemini-generated summary of the post, key themes, and research gaps related to the topics of the post. Each generated insight is stored and cached in the database for 7 days to reduce AI consumption.

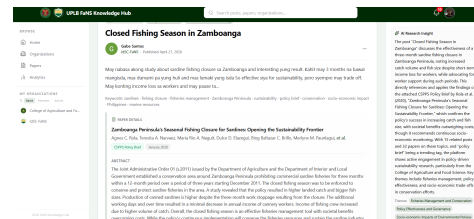


Fig. 7. Post Page with AI Related Insights

- 5) **Other AI Enhancements:** Other AI features are used for the ease of use of content curation, and accuracy of papers uploaded. Each post is automatically categorized into eight different subtopics:

- **Food Security**
- **Nutrition & Health**
- **Food Science & Technology**
- **Agricultural Systems**
- **Policy & Governance**
- **Traditional Knowledge**
- **Education & Communication**
- **Environment & Sustainability**

This is to assign topics and categorize posts depending on their given topic. Metadata extraction is also done to every research paper being uploaded in the system. This is done through research paper posts by users in their organizations. At the same time, in bulk upload of research papers by organization admins, each paper is extracted for its metadata to be added to the MongoDB and Elasticsearch. Duplicates are screened and skipped in the uploading.

D. Data Collection

a) **User-Driven PDF Upload:** The system is designed for decentralized contribution by IdSC-FaNS members and UPLB constituents rather than bulk ingestion by the researcher. A user uploads a PDF through the Submit a Post dialog, the file is then sent as a request to `POST /api/upload/paper`, and stored in DigitalOcean Spaces. The PDF is encoded and sent to Google Gemini 2.5 Flash with a prompt instructing it to return the metadata of the paper. The extracted metadata prefills the create paper form, which the user reviews, edits, and submits. This is a **human-in-the-loop** design consistent with Human-Centered AI principles [15].

Fig. 8. AI Extraction — Post Submission

b) **Admin Pre-seeded Research:** For demonstration and evaluation purposes, publicly accessible research provided by the UPLB University Library in the initial data collection phase, and manually chosen papers from the University Knowledge Digital Repository (UKDR), were added to the system through the bulk upload feature as an organization admin. The collection of data involves the gathering of relevant material and making sure the data is structured, formatted, and indexed properly in Elasticsearch.

Research submitted by the testers and evaluators were also added to the system.

E. Testing

A **convergent mixed-methods evaluation** [16] was adopted, combining quantitative usability and task-success measurements with qualitative thematic feedback collected in the same session. The evaluation operationalizes usability per the **ISO 9241-11:2018** definition as a composite of effectiveness, efficiency, and satisfaction [17].

1) **Participants and Sampling:** The evaluation was administered as a fully remote, unmoderated study [18], [19]. Each participant was given a link to the deployed website and the Google Form and worked through the test at their own pace and on their own machine.

Three roles were created through **role-based form branching**. A screening question (“What role were you assigned for this evaluation?”) routed each participant into one of the three task blocks — Track A (Standard User), Track B-Org (Organization Administrator), and Track B-Site (Website Administrator). The tracks were not randomly assigned; instead, the researcher assigned each participant their task depending on their role in IdSC-FaNS and their relevant qualifications.

The target total was $N \geq 15$, the working floor for stable SUS confidence intervals [20]. Nielsen’s five-user heuristic [21] and Faulkner’s refinement [22] indicate that 10–15 testers detect $\geq 95\%$ of usability problems, supporting this floor. A total of $N = 26$ responses were collected, distributed as **20 standard users, 5 organization administrators, and 1 website administrator**. The total number of valid responses comfortably exceeds the $N \geq 15$ stable-SUS floor, and Track A and Track B-Org each meet or exceed the five-user range at which single-role usability findings stabilize [21], [22]. In contrast, Track B-Site did not meet the sample threshold: only one website administrator was available within the evaluation window because the role is provisioned for at most a handful of accounts in the production deployment. Site-admin findings are therefore reported as a **single-case feasibility observation** confirming that the site-administrator workflow is operable end-to-end and are excluded from role-stratified statistical inference; the eligible site-administrator population is itself small (on the order of three accounts in the production deployment), and the inferential follow-up is named in Chapter V.

A second sample-related limitation surfaced after the testing and is recorded here. One Track B-Org participant was assigned the org-administrator role for the evaluation but was not actually elevated to org-admin permissions in the production system. The participant’s open-ended response described an inability to locate admin controls on the organization page, which the rest of the Track B-Org task data did not align with. Because the system enforces role-based access control, the participant’s failures on admin-gated tasks reflect the absence of the admin role rather than a usability property of the admin tools. The participant is therefore excluded from Track B-Org task and SUS analyses, dropping the role denominator for Track B-Org to $n = 4$ for those analyses; the participant’s responses on items that do not depend on admin privileges (pooled SUS, demographics, AI insight Likerts, AI extraction accuracy, and the public-analytics task) are retained. The reduced denominator is one participant below the five-user floor cited in Nielsen [21] and refined by Faulkner [22]. Track B-Org is therefore reported with sample-based descriptive statistics only, while the inferential SUS claim is carried by the pooled $N = 26$ which exceeds the $N \geq 15$ stable-CI floor cited from Lewis [20]. The case is itself an argument for a synchronous provisioning verification step in any follow-up evaluation, which is discussed in Chapter V.

2) *Evaluation Instruments*: SUS remains one of the most widely used tools for measuring usability in HCI because it is short, free, and still dependable even with smaller sample sizes. It also gives a clear 0–100 score that can be interpreted using familiar labels, such as “Good” (around 70–84) and “Excellent” (85 and above) [23]. Task success was also included to measure effectiveness as stated in ISO 9241-11:2018, [17]. Along with that, Single Ease Question (SEQ) test was also employed to capture perceived task difficulty in a single question making it more practical for the researcher and the evaluation participants alike [24]. Task success and SEQ are done for every task in supplementary with each other to determine the completeness, accuracy, and difficulty each task is. Feature-level Likert ratings were added so the Results chapter can clearly show which areas performed well or need improvement (search, AI, community, analytics, and admin tools) [25]. For AI insights, ratings were completed by domain experts since judging summary quality and research-gap validity requires subject knowledge [15]. Extraction accuracy was evaluated using a per-field information-retrieval approach [26]. Finally, open-ended responses were analyzed using Braun and Clarke’s six-phase thematic analysis to generate evidence-rich insights for the Discussion chapter [27].

The table of all evaluation instruments used is found on the next page. Seven instruments were used. A mix of both qualitative and quantitative methods are optimal to gather complex but relevant data. Subjective and objective methods were also used to accurately test the features of the system while obtaining anecdotal experiences when using the system.

TABLE IV
EVALUATION INSTRUMENTS

ID	Instrument	Data type	Scale	Citation
I-1	System Usability Scale	Quantitative	10 items, 1–5 Likert, 0–100	[23], [28]
I-2	Task success rate	Quantitative	Yes / Partial / No per task	[17], [29]
I-3	Single Ease Question (SEQ)	Quantitative	1–5 per task	[24], [30]
I-4	Feature-specific satisfaction	Quantitative	5-point Likert grids	[25], [31]
I-5	AI insight accuracy	Mixed	4 × 5-point Likert per insight	[15], [32]
I-6	AI extraction accuracy	Quantitative	Per-field Correct / Partial / Incorrect / N/A	[26]
I-7	Open-ended thematic feedback	Qualitative	6 free-text items, inductive coding	[27]

The test was administered using Google Forms. Evaluators were instructed to have both the form and the website open in tabs side-by-side. They were to strictly follow the given instructions per task before answering the relevant questions. Because the evaluation was self-paced and unmoderated, the protocol is realized as the structure of the Google Form rather than a timed in-person session. The form is a single linear flow: consent and Data Privacy Act notice → demographics (age band, gender, role/affiliation, qualifications, research-platform usage frequency, device used) → screening item (“What role were you assigned for this evaluation?”) → role-branched task block → AI-evaluation block (shared) → public-analytics

task (shared) → SUS (10 items) → feature-satisfaction Likert grids (Search and Papers; AI Features; Community and Organization; Analytics; Admin Tools — answered only by admin tracks; Overall) → six open-ended thematic items → submission. Per-task instruments — success outcome (Yes / Partially / No), SEQ difficulty (1–5, where 1 = Very Easy and 5 = Very Difficult), and an optional free-text problem report — are placed immediately below each task instruction so that the rating reflects the just-completed task.

The individual sections of the form are discussed more thoroughly in the Results and Discussion. It is also important to note that for testing, all new accounts were immediately added to the *IdSC-FaNS* organization as members to ease the bottleneck of having to approve the users in real-time.

IV. RESULTS AND DISCUSSION

This chapter reports the results of the evaluation defined in Chapter III. It interprets the data collected against the five research objectives stated in Chapter I. The sample data comprises $N = 26$ participants: 20 standard users, 5 organization administrators, and 1 website (super) administrator. This chapter reports the participant’s profile, the data and results from the System Usability Scale, per-task success and effort, feature-specific satisfaction, AI-assisted feature evaluation, and the qualitative evaluation in that order.

A. Participant Profile

The 26 participants spanned four age bands and two affiliations of interest. Twelve (46%) were 18–24, nine (35%) were 25–34, four (15%) were 45–54, and one (4%) was 55+. Eighteen (69%) identified as female and eight (31%) as male. Reported affiliation was distributed across “Other” (15; 58%), Undergraduate Student (4; 15%), Faculty (3; 12%), Research Staff (3; 12%), and Administrative Staff (1; 4%). Thirteen participants listed an additional research qualification as Registered Nutritionist-Dietitian and a further five listed graduate or professional credentials (PhD, MSc ×2, Registered Agriculturist, DMD); the remaining seven reported none. Research platform usage frequency of the users are around the higher end of 1–5 scale. Ten participants report 3, eight reported 4, four report 5, and four reported 2; no participant reported 1. Twenty-five participants (96%) tested on a laptop or desktop and one used a smartphone.

All but one of the participants who answered “Other” in their UPLB role are UPLB Alumni who graduated with a Degree in BS Nutrition under the Institute of Human Nutrition and Food.

Crossing affiliation against assigned evaluation role shows that the five organization administrators were drawn from “Other” (4) and Undergraduate Student (1), and the single website administrator was drawn from “Other”; the standard-user track included all three Faculty respondents and all three Research Staff respondents. This distribution is consistent with the role-provisioning approach described in Chapter III: admin accounts were elevated by the researcher before recruitment and so the admin tracks reflect *who was given an admin account*, not *who happened to be senior in real life*. As noted

TABLE V
FORM STRUCTURE (REMOTE UNMODERATED, ROLE-BRANCHED)

Section	Items	Instruments	Track
Consent & demographics	7 items	—	All
Screening (assigned role)	1 item	—	All
Track A task block	7 task groups (sign-in & profile, profile expertise, search, save/cite/download, follow & join org, like/comment, submit post)	I-2, I-3	Track A
Track B-Org task block	8 task groups (approve, reject, manage join requests, pin, promote, moderate report, bulk upload, org analytics)	I-2, I-3	Track B-Org
Track B-Site task block	5 task groups (approve org request, site analytics, announcement, user management, feature post)	I-2, I-3	Track B-Site
AI-evaluation block	Insight 1 (mandatory) + Insight 2 (optional) + paper upload	I-2, I-3, I-5, I-6	All tracks
Public-analytics task	1 task	I-2, I-3	All tracks
SUS	10 items	I-1	All tracks
Feature-satisfaction Likert grids	22 items across 6 grids (Admin grid answered only by admins)	I-4	All tracks
Open-ended	6 items (most useful, most confusing, missing features, bugs, plan to use again, other comments)	I-7	All tracks

TABLE VI
REALIZED PARTICIPANT DEMOGRAPHICS ($N = 26$)

Variable	Level	Count	%
Age band	18–24	12	46
	25–34	9	35
	45–54	4	15
	55+	1	4
Gender (assigned at birth)	Female	18	69
	Male	8	31
UPLB role / affiliation	Other	15	58
	Undergraduate Student	4	15
	Faculty	3	12
	Research Staff	3	12
	Administrative Staff	1	4
Additional qualifications (free-text)	RND	13	50
	PhD	2	8
	MS / MSc	2	8
	R.Agr. / DMD	2	8
	None / N/A	7	27
Research-platform usage (1 rare – 5 daily)	3	10	38
	4	8	31
	5	4	15
	2	4	15
Device	Laptop / Desktop	25	96
	Smartphone	1	4
Assigned evaluation role	Standard User	20	77
	Organization Admin	5	19
	Website / Super Admin	1	4

in Chapter III, one Track B-Org participant was assigned the org-administrator role but was not elevated in the production system, and is excluded from admin-gated analyses for the reasons given there. The Track B-Org admin-track denominator is therefore $n = 4$ for SUS and per-task analyses; the participant’s responses are retained in the pooled SUS and in analyses not contingent on admin provisioning (AI insight Likerts, AI extraction accuracy, public-analytics task).

B. System Usability Scale

The pooled SUS score across all 26 participants was **mean** = 83.17, **SD** = 17.61, **median** = 92.5, **range** 50.0–100.0. The 95% bootstrap CI (10,000 resamples) was [76.35, 89.52]. The mean falls in the “**Excellent**” band on the Bangor adjective scale (≥ 85 in Bangor’s strict cut, “Good” between 72 and 85) [23]. Even the lower bound of the bootstrap CI sits inside the “Good” band. Sixteen of twenty-six participants (62%) scored in “Excellent” (≥ 85), three (12%) in “Good” (72–84), five (19%) in “OK” (52–71), and two (8%) in “Poor” (38–51). No participant scored below 50.

By track, **Track A** ($n = 20$) averaged 82.75 (SD = 18.44, median 92.5, range 50–100); **Track B-Org** ($n = 4$, **after excluding the unprovisioned participant per Chapter III**) averaged 89.4 (SD ≈ 6.6 , median ≈ 91.25 , range ≈ 80 –95); and **Track B-Site** ($n = 1$) scored 95.0. After the provisioning-related exclusion, the role-stratified Track B-Org SUS rises into the Good–Excellent band, which is consistent with Track A’s distribution and supports the claim that the admin-side workflows do not depress perceived usability when admin permissions are correctly provisioned. P21’s SUS score is retained in the pooled grid because SUS does not depend on admin permissions; only the Track B-Org role-stratified line excludes it. The website administrator’s single score is reported descriptively per the limitation in Chapter III as a single-case feasibility observation and is not used for statistical inference.

C. Task Success and Effort

Per-task success was coded ($Yes + 0.5 \cdot Partial$)/total · 100 [33]. Effort was reported on the form’s 1–5 SEQ scale where

TABLE VII
SUS RESULTS (POOLED AND ROLE-STRATIFIED)

Stratum	n	Mean	SD	Median	Range	95% Bootstrap CI	Adjective band
Pooled	26	83.17	17.61	92.5	50.0–100.0	[76.35, 89.52]	Good–Excellent
Track A (Standard User)	20	82.75	18.44	92.5	50.0–100.0	—	Good–Excellent
Track B-Org (Org Admin)	4	89.4	6.6	91.25	80.0–95.0	—	Good–Excellent
Track B-Site (Site Admin)	1	95.00	—	—	—	—	Excellent ($n = 1$, single-case)

1 = **Very Easy** and 5 = **Very Difficult**, so lower means easier. Tasks restricted to a track use that track’s n as the denominator; the AI-evaluation, paper-upload, and public-analytics tasks were given to all tracks and use $n = 26$. Track B-Org tasks use $n = 4$ following the exclusion of the unprovisioned participant identified in Chapter III; the original responses for that participant on admin-gated tasks reflect the absence of the org-admin role rather than a usability property of the admin tools and are therefore not included in the per-task denominators below.

Twenty-two of the twenty-three tasks scored 95% or higher on success and at or below 1.5 on SEQ. Two tasks remain below the threshold after the Track B-Org exclusion:

- **A4 (Save paper / cite / download)** — 87.5% success (15 Yes, 5 Partial), mean SEQ 1.25. The qualitative reports show this is from the cite-button discoverability problem, not an actual failure of save or download.
- **A7 (Submit a discussion post)** — 80.0% success (16 Yes, 0 Partial, 4 No), mean SEQ 1.40. Reports describe an “internal server error” on submit. One participant found that adding a text body to the post body field let the submission go through, which points to a server-side validation rule that the client does not enforce.

After excluding the unprovisioned Track B-Org participant, every B-Org task with a sample-based denominator returns to or near 100% success. B-Org tasks 3, 4, 5, 6, and 8 (manage join requests; pin; promote; moderate report; view org analytics) each scored 100% (4/4), with mean SEQ values that drop below the 1.5 threshold once the unprovisioned participant’s high-effort responses are excluded. B-Org 7 (bulk upload) is reported with the corrected denominator in Table VIII; the residual partial responses on that task are attributable to “some files failed” and “duplicate” notices, both of which are correct system behaviour rather than usability faults. With this correction, no task with a sample-based denominator crosses the 1.5 SEQ threshold or the 3.0 difficulty threshold flagged in the data-analysis plan.

Branched yes/no items confirm the same picture. Of the eighteen Track A participants who reached the post-submission confirmation, eighteen rated “Yes, very clear” that the post had been submitted *for review* rather than published immediately, and two answered “I didn’t notice”. This shows that Objective 5’s approval pipeline is communicated correctly when the submit actually succeeds. Sixteen of twenty Track A search responses described the search results as “Yes, very relevant”, three as “Somewhat relevant”, and one as “Not relevant”.

D. Feature-Specific Satisfaction

Five Likert grids and one Overall grid were given (1–5 Likert, 5 = Strongly Agree). Each item had $n = 26$. The Admin Tools grid was answered by all participants, but the instructions only asked the admin tracks to give a substantive rating, so its means must be read with that in mind. The $n = 26$ distribution is bimodal (12 “Strongly Agree”, 11 “Neutral”, 3 “Agree” on item 121), which is what happens when non-admins default to “Neutral”.

A few things are worth pointing out. Every grid except AI Features and Admin Tools sits at or above 4.46. The **AI-trust item** (3.81 ± 0.94) is the only sub-4 item in the entire instrument, which lines up with the qualitative theme that AI insights are valued but not blindly trusted. The Overall grid is the strongest of the six: “valuable for the UPLB research community” reaches 4.92 with the lowest dispersion ($SD = 0.27$), which supports the system-level claim even with the AI-trust dip.

E. AI-Assisted Feature Evaluation

Two AI insights were rated by every participant (the second insight had been an “if time” item but the data shows all 26 responded). Each insight was rated on four dimensions: summary accuracy, theme correctness, gap validity, and usefulness. Means landed between 3.88 and 4.12 with fairly high SDs (1.21–1.31). The high SDs come from a small minority of participants who rated low.

The spread here is what’s worth looking at. A subset of participants rated the AI insights “1” on summary accuracy, which lines up with two of the qualitative responses. One participant wrote that “some AI summary [is] not mentioned in the [post]” and guessed the model was “hallucinating”, and another wrote “Just not reliant on AI doing my research”. Since the AI-insight evaluation is a domain judgment about food and nutrition security research [15], this kind of spread is expected. Domain experts disagree about the model’s gap claims, and the system’s response to that disagreement (the “Was this useful?” feedback control on each insight) becomes a feature that can be evaluated in any future iteration.

For paper-metadata extraction, every field used a per-field correctness category with an explicit “N/A” option to mark “I couldn’t see the metadata”. Three of the 26 participants reported they could not see the extracted metadata after upload (Track A: row 10 “File was successfully uploaded but was not able to see extracted metadata”). These responses are shown as “N/A” in the per-field counts and are excluded from the percentage denominators below. They are reported separately

TABLE VIII

TASK SUCCESS AND SEQ BY TASK. SEQ IS REPORTED ON THE FORM'S 1 = VERY EASY, 5 = VERY DIFFICULT SCALE; LOWER MEANS INDICATE EASIER TASKS.

Task ID	Description	Track(s)	n	Success %	Y / P / N	Mean SEQ
A1	Sign-in & report papers count	A	20	100.0	20 / 0 / 0	1.20
A2	Update profile expertise	A	20	100.0	20 / 0 / 0	1.00
A3	Search for a paper	A	20	100.0	20 / 0 / 0	1.05
A4	Save / cite / download	A	20	87.5	15 / 5 / 0	1.25
A5	Follow & join organization	A	20	100.0	20 / 0 / 0	1.05
A6	Like or comment	A	20	97.5	19 / 1 / 0	1.10
A7	Submit a discussion post	A	20	80.0	16 / 0 / 4	1.40
Shared-1	Read AI insight (Insight 1)	All	26	100.0	26 / 0 / 0	—
Shared-2	Upload a PDF and review extracted meta-data	All	26	88.5	23 / 0 / 3	1.27 (overall)
Shared-3	View public analytics	All	26	100.0	26 / 0 / 0	1.27 (overall)
B-Org 1	Approve a pending post	B-Org	4	100.0	4 / 0 / 0	1.00
B-Org 2	Reject a pending post	B-Org	4	100.0	4 / 0 / 0	1.20
B-Org 3	Manage a join request	B-Org	4	100.0	4 / 0 / 0	1.25
B-Org 4	Pin a post	B-Org	4	100.0	4 / 0 / 0	1.50
B-Org 5	Promote a member	B-Org	4	100.0	4 / 0 / 0	1.50
B-Org 6	Moderate a reported post	B-Org	4	100.0	4 / 0 / 0	1.50
B-Org 7	Bulk upload a folder	B-Org	4	87.5	3 Y, 1 P (some failed / duplicates)	1.25
B-Org 8	View organization analytics	B-Org	4	100.0	4 / 0 / 0	1.25
B-Site 1	Approve organization-creation request	B-Site	1	100.0	1 / 0 / 0	1.00
B-Site 2	Site analytics	B-Site	1	100.0	1 / 0 / 0	1.00
B-Site 3	Site-wide announcement	B-Site	1	100.0	1 / 0 / 0	2.00
B-Site 4	User management	B-Site	1	100.0	1 / 0 / 0	1.00
B-Site 5	Feature a post	B-Site	1	100.0	1 / 0 / 0	1.00

TABLE IX

FEATURE-SATISFACTION ITEMS, MEAN (SD) ON A 1–5 LIKERT (5 = STRONGLY AGREE). EACH ITEM HAS $n = 26$; THE ADMIN TOOLS GRID WAS ANSWERED SUBSTANTIVELY BY ADMIN TRACKS ONLY ($n = 6$).

Cluster	Item	Mean	SD
Search and Papers	Search helped me find relevant research	4.88	0.33
	Filters and sorting were useful	4.85	0.46
	Citation generator is useful	4.69	0.68
	PDF download was fast and straightforward	4.96	0.20
AI Features	Metadata extraction would save time vs. manual entry	4.77	0.59
	AI research insights are accurate and valuable	4.62	0.64
	Research gaps identified are worth investigating	4.50	0.65
	I would trust AI features in my own research workflow	3.81	0.94
Community and Organization	Post and comment system supports academic discussion	4.88	0.33
	Organization structure is clear and well-organized	4.69	0.55
	Approval workflow ensures quality without burden	4.69	0.62
	Notifications kept me informed	4.46	0.81
Analytics	Charts help understand current state of research	4.88	0.43
	Analytics useful for identifying research priorities	4.88	0.43
Admin Tools (admins, $n = 6$)	Org admin tools are easy to use	4.04	0.96
	Site admin dashboard provides useful oversight	4.12	0.99
	Bulk paper upload is practical for building libraries	4.00	0.94
	Role-based permission model feels appropriate	4.08	0.98
Overall	I can see myself / my org using this regularly	4.77	0.51
	This system would be valuable for the UPLB research community	4.92	0.27
	The website feels reliable and responsive	4.88	0.33
	The visual design is clean and professional	4.92	0.27

TABLE X

AI INSIGHT LIKERTS (5 = STRONGLY AGREE, $n = 26$ PER INSIGHT)

Dimension	Insight 1 mean (SD)	Insight 2 mean (SD)
Summary accuracy	3.88 (1.28)	4.00 (1.26)
Theme correctness	4.00 (1.30)	4.04 (1.28)
Gap validity	4.12 (1.21)	4.04 (1.25)
Usefulness	4.12 (1.31)	4.00 (1.26)

as a $3/26 = 11.5\%$ upload-rendering failure rate, which fed the 88.5% task-success rate for Shared-2.

TABLE XI

AI EXTRACTION ACCURACY PER FIELD. WILSON 95% CI ON THE N/A-EXCLUDED DENOMINATOR; THREE OF 26 PARTICIPANTS REPORTED THEY COULD NOT SEE THE EXTRACTED METADATA AFTER UPLOAD AND ARE EXCLUDED FROM THE PER-FIELD DENOMINATOR.

Field	Correct / n	% Correct	Wilson 95% CI
Title	23 / 23	100.0	[85.7, 100.0]
Authors	22 / 23	95.7	[79.0, 99.2]
Abstract	23 / 23	100.0	[85.7, 100.0]
Keywords (all relevant)	21 / 23	91.3	[73.2, 97.6]
Year / Journal (all correct)	19 / 22	86.4	[66.7, 95.3]

The single Authors miss was rated “Partially Correct”. On Year / Journal, one paper had no journal field in the source PDF (correctly flagged as “Not present in the paper”), one was rated “All incorrect”, and one was rated “Some correct”. The aggregate “Overall, the AI metadata extraction was accurate enough to be useful without major manual corrections” Likert came back at **mean** = 4.54, **SD** = 0.81 ($n = 26$). This matches the per-field numbers and supports the human-in-the-loop framing of the extraction pipeline in Chapter III [15].

V. CONCLUSION AND RECOMMENDATIONS

A. Conclusion

This study set out to design and develop UPLB IdSC-FaNS Knowledge HUB: A Knowledge Management System (KMS) for food and nutrition security research in the Philippines, built specifically for the UPLB Interdisciplinary Studies Center on Food and Nutrition Security (IdSC-FaNS). The system was developed as a web application using a hybrid MongoDB and Elasticsearch architecture, with AI-assisted features for paper metadata extraction and research insights, and a community layer for posts, comments, and organization and site level moderation. The evaluation in Chapter IV used a fully remote, unmoderated form on $N = 26$ participants drawn from the UPLB community across three tracks: 20 standard users, 5 organization administrators, and 1 website administrator.

The pooled SUS score of 83.17 (95% CI [76.35, 89.52], median 92.5) places the system in the Good–Excellent band of the Bangor adjective scale. Twenty of twenty-three tasks scored $\geq 95\%$ success and ≤ 1.5 SEQ on a 1 = Very Easy /

5 = Very Difficult scale. AI metadata extraction was the most-named useful feature in the open-ended responses, and field-level extraction accuracy ranged from 86.4% on Year / Journal to 100% on Title and Abstract. The five research objectives stated in Chapter I are revisited below against the data from Chapter IV.

Objective 1 — Provide a centralized platform for IdSC-FaNS research. This objective is met. The four Search and Papers Likerts landed between 4.69 and 4.96, the public-analytics task succeeded for all 26 participants, and the qualitative responses (theme T7) collected unprompted system level endorsements naming the hub as something participants wished had existed during their own research. The main bug that challenged this objective is the cite button disappearance. A straightforward fix is expected and does not threaten the structure and integrity of the system.

Objective 2 — Enable advanced search and discovery using Elasticsearch. This objective is met without issue. Every Track A search task scored 100% success with SEQ ≤ 1.05 , and sixteen of twenty participants rated the search results as “very relevant”. The two filter regressions in theme T5 (oldest-first ordering and the year-range filter) are scoped, in-scope fixes and do not put the BM25-and-aggregation approach into question.

Objective 3 — Provide data visualization and analytics tools. This objective is met without issue. Both Analytics Likerts hit 4.88, the public-analytics task succeeded for all 26 participants, and no participant raised concerns about the chart set in the open-ended items.

Objective 4 — Support community-driven curation through organizations and moderated discussions. This objective is met. The four Community and Organization Likerts landed between 4.46 and 4.88, the like-or-comment task succeeded for 19 of 20 participants, and all eighteen Track A participants who reached the post-submission confirmation rated “Yes, very clear” that the post had been submitted *for review* rather than published immediately. After the provisioning-related Track B-Org exclusion described in Chapter III, every admin-gated B-Org task (3 through 6 and 8) returns to 100% success on the corrected $n = 4$ denominator, indicating that the admin-side moderation workflows are usable when admin permissions are correctly provisioned. The post-submit server error (theme T4) is a scoped fix and does not put the organization-and-moderation curation model into question.

Objective 5 — Provide AI-assisted curation through automated metadata extraction and research insight generation. This objective is met. AI metadata extraction was the most-named useful feature in the open-ended responses, per-field extraction accuracy ranged from 86.4% on Year / Journal to 100% on Title and Abstract, and the aggregate “accurate enough to be useful without major manual corrections” item came back at 4.54 ± 0.81 . The AI-trust item (3.81 ± 0.94) is the only sub-4 item in the entire feature-satisfaction instrument and lines up with theme T2’s “hallucination” concern; the fix is provenance and verification cues rather than a pipeline redesign.

Across all five objectives, the system meets each one at both the satisfaction and task-success layers. What remains is a bug-

fix list (cite-button affordance, post-body server validation, and filter regressions), not a structural rework. The most consistent qualitative finding across affiliations and tracks is that participants would use the system again: twelve out of twenty-two substantive responses planned to explore further and zero declined outright.

B. Recommendations

The recommendations below are split into two groups. The first set covers features and fixes to the system itself, drawn from the bugs and feature requests named directly by participants in the open-ended items. The second set covers recommendations for future studies on the system.

1) Recommendations for Features and Fixes to the System:

- 1) **Restore the cite-button affordance on the paper view** Five participants reported the cite button was missing or hard to find on Task A4, and the 87.5% success rate on that task is fully explained by this. The control should be surfaced on the paper view next to the existing copy-link, save, and download controls.
- 2) **Fix the post-submit “internal server error”** Three to four participants hit an internal server error on Task A7. P5 identified the workaround as adding a text body to the post body field, which means the server is enforcing a non-empty body field that the client does not enforce. The fix is to either enforce the body field on the client or relax the server validation, depending on which behavior is intended.
- 3) **Fix the oldest-first / newest-first sort and the year-range filter** P17 reported that the “Oldest first” sort did not return the oldest paper first, and P21 reported that a 2025–2025 year-range query returned posts from 2026. These reports concern the public papers page and are independent of the provisioning issue noted in Chapter III; they are likely syntax inaccuracies in the backend query construction and are an easy fix.
- 4) **Investigate the metadata-not-rendered-after-upload case** Three of 26 participants reported that the upload succeeded but the extracted metadata was not visible. This is the 11.5% upload-rendering failure rate that drove the 88.5% task-success rate on Shared-2. The cause should be checked at the upload pipeline and the post-upload paper-view render.
- 5) **Add insight proofs and verification cues** The AI-trust item (3.81 ± 0.94) was the only sub-4 item in the entire feature-satisfaction instrument, and the qualitative responses (theme T2) named “hallucination” as the concern. Surfacing the source posts or papers behind each insight, and showing which parts of the insight came from which source, would address this without changing the underlying extraction or insight pipelines. Altering the Gemini prompt to a more appropriate and more encompassing prompt may fix this issue.

2) Recommendations for the Study:

- 1) **Re-test the org-join-approval and admin-elevation flows end-to-end.** Auto-membership in IdSC-FaNS and

pre-elevated admin privileges short-circuited these two real-world workflows during the evaluation. A future evaluation should test them end-to-end from a freshly registered standard-user account.

- 2) **Run a follow-up evaluation with a larger admin sample.** Track B-Org rests on $n = 5$ and Track B-Site is a single observation. A follow-up evaluation with more organization administrators and at least one additional website administrator would tighten the role-stratified SUS estimates and let the admin-tools cluster (currently 4.00–4.12) be evaluated on a sample-based denominator rather than a mixed $n = 26$ grid.
- 3) **Expand the data ingestion pipeline beyond the initial UPLB Library dataset** The current corpus relies on the metadata and DOI-based enrichment of the initial UPLB Library export. Connecting the ingestion pipeline to additional sources (other UP units, partner institutions, and open-access repositories like FAO) would strengthen the centralized-platform claim over time.

Overall, a moderated evaluation session may be more appropriate to more extensively test all workflows. The current setup of a remote, unmoderated evaluation showed weaknesses in its capability to evaluate admin approval workflows: a moderator would also have caught the missing org-admin elevation on the affected Track B-Org account at the start of the session, before any task data was collected, rather than after the fact through the open-ended responses. A synchronous provisioning-verification step is therefore an explicit requirement for any follow-up evaluation. Expanding the realized samples for Track B-Org and especially Track B-Site would also be beneficial in testing the admin controls present in the system. It is important to note that none of the recommendations require a system architecture and workflow overhaul. Simple polishing and streamlining would be enough to address the qualitative comments and feedback and should reflect positively in the quantitative assessments.

C. Closing Remarks

The UPLB IdSC-FaNS Knowledge Hub meets the five research objectives stated in Chapter I at both the satisfaction and task-success layers, with a focused bug-fix list rather than a structural rework. The AI metadata extraction was the most-named useful feature in the open-ended responses, the search and analytics features cluster near the ceiling on the Likert grids, and the community layer was endorsed as a tool participants would use again. The strongest single qualitative claim, repeated across affiliations and tracks is that the system would be valuable for the UPLB research community, with several participants stating they wished the platform had existed during their own research. The recommendations listed above are the next concrete steps for taking the system from a thesis prototype to a production tool that IdSC-FaNS can hand to its researchers, students, and policymakers.

REFERENCES

- [1] I. Angeles-Agdeppa *et al.*, “Food sources, energy and nutrient intakes of adults: 2013 Philippines National Nutrition Survey,” *Nutrition Journal*, vol. 18, no. 1, Oct. 2019.

- [2] F. L. Durano, J. M. Sarmiento, G. Dee, G. D. Wahing, A. M. Traje, and D. Baker, "Food security of agri-food system actors during the COVID-19 pandemic in the Philippines: Post-pandemic implications to sustainable development," *Journal of Agriculture and Food Research*, vol. 18, p. 101284, Dec. 2024.
- [3] K. Heitman, J. Hubbard, L. Easter, and J. Kilkus, "Looking to the future: Agendas, directions, and resources for nutrition research," *Nutrition in Clinical Practice*, Apr. 2024.
- [4] Global Nutrition Report, "Philippines nutrition profile," <https://globalnutritionreport.org/resources/nutrition-profiles/asia/south-eastern-asia/philippines/>, Feb. 2020.
- [5] M. S. Righettini and E. Bordin, "Exploring food security as a multidimensional topic: twenty years of scientific publications and recent developments," *Quality & Quantity*, Aug. 2022.
- [6] A. S. Van Zyl *et al.*, "Knowledge management and organizational effectiveness," *Journal of Knowledge Management Practice*, 2014.
- [7] IBM, "What is knowledge management?" <https://www.ibm.com/topics/knowledge-management>, 2023.
- [8] A. Maharani *et al.*, "Knowledge management system design using the People, Process, Technology framework," *International Journal of Engineering Research and Technology*, 2020.
- [9] A. Usman *et al.*, "Components of knowledge management systems in modern organizations," *Journal of Information and Knowledge Management*, 2019.
- [10] Harvard Library, "DASH: Digital access to scholarship at Harvard," <https://dash.harvard.edu/>, 2024.
- [11] Food and Agriculture Organization, "Knowledge sharing," <https://www.fao.org/knowledge/en/>, 2023.
- [12] M. S. Vinutha and M. C. Padma, "A survey of AI and ML techniques for search engines," *International Journal of Computer Applications*, 2020.
- [13] S. Robertson and H. Zaragoza, *The Probabilistic Relevance Framework: BM25 and Beyond*, ser. Foundations and Trends in Information Retrieval. Now Publishers, 2009.
- [14] M. McCandless, E. Hatcher, and O. Gospodnetić, *Lucene in Action*, 2nd ed. Manning Publications, 2010.
- [15] B. Shneiderman, "Human-centered artificial intelligence: Reliable, safe and trustworthy," *International Journal of Human-Computer Interaction*, vol. 36, no. 6, pp. 495–504, 2020.
- [16] J. W. Creswell and V. L. Plano Clark, *Designing and Conducting Mixed Methods Research*, 3rd ed. SAGE Publications, 2017.
- [17] International Organization for Standardization, "ISO 9241-11:2018 ergonomics of human-system interaction — part 11: Usability: Definitions and concepts," Standard, Geneva, Switzerland, 2018.
- [18] J. Sauro and J. R. Lewis, *Quantifying the User Experience: Practical Statistics for User Research*. Morgan Kaufmann, 2012.
- [19] W. Albert and T. Tullis, *Measuring the User Experience: Collecting, Analyzing, and Presenting Usability Metrics*, 2nd ed. Morgan Kaufmann, 2013.
- [20] J. R. Lewis, "The System Usability Scale: Past, present, and future," *International Journal of Human-Computer Interaction*, vol. 34, no. 7, pp. 577–590, 2018.
- [21] J. Nielsen, "Why you only need to test with 5 users," <https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/>, 2000.
- [22] L. Faulkner, "Beyond the five-user assumption: Benefits of increased sample sizes in usability testing," *Behavior Research Methods, Instruments, & Computers*, vol. 35, no. 3, pp. 379–383, 2003.
- [23] A. Bangor, P. Kortum, and J. Miller, "Determining what individual SUS scores mean: Adding an adjective rating scale," *Journal of Usability Studies*, vol. 4, no. 3, pp. 114–123, 2009.
- [24] J. Sauro and J. S. Dumas, "Comparison of three one-question, post-task usability questionnaires," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI)*, 2009, pp. 1599–1608.
- [25] J. R. Lewis and J. Sauro, "Item benchmarks for the System Usability Scale," *Journal of Usability Studies*, vol. 13, no. 3, pp. 158–167, 2018.
- [26] G. Hripcsak and A. S. Rothschild, "Agreement, the f-measure, and reliability in information retrieval," *Journal of the American Medical Informatics Association*, vol. 12, no. 3, pp. 296–298, 2005.
- [27] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77–101, 2006.
- [28] J. Brooke, "SUS: A quick and dirty usability scale," in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and A. L. McClelland, Eds. Taylor and Francis, 1996, pp. 189–194.
- [29] J. Rubin and D. Chisnell, *Handbook of Usability Testing: How to Plan, Design, and Conduct Effective Tests*, 2nd ed. Wiley, 2008.
- [30] J. Sauro, "SEQ: 10 reasons to use the Single Ease Question," <https://measuringu.com/seq10/>, 2012.
- [31] R. Likert, "A technique for the measurement of attitudes," *Archives of Psychology*, vol. 22, no. 140, pp. 1–55, 1932.
- [32] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," *arXiv preprint arXiv:1702.08608*, 2017.
- [33] J. Sauro, *A Practical Guide to Measuring Usability: 72 Answers to the Most Common Questions about Quantifying the Usability of Websites and Software*. Measuring Usability LLC, 2011.